CHAPTER 11

DUAL NATURE OF RADIATION AND MATTER

CHAPTER ESSENCE AT A GLANCE:
Experimental evidence for the particle nature of light through the photoelectric effect, Einstein's equation, and de Broglie matter waves.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Photoelectric Emission & Photons:** Incident photons transfer their entire quantum of energy instantaneously to individual surface electrons to cause emission.

• **Work Function & Threshold Frequency:** Emission cannot occur unless incident photon frequency exceeds a characteristic threshold frequency providing minimum binding energy.

• **Intensity vs Frequency Independence:** Light intensity controls photocurrent, while incident frequency exclusively determines maximum kinetic energy and stopping potential of photoelectrons.

• **de Broglie Matter Wave Hypothesis:** Moving material particles exhibit dual wave-particle properties with a matter wavelength inversely proportional to linear momentum.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$K_{\max} = h\nu - \Phi_0 = eV_0$$

Einstein's Photoelectric Equation

$$\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{h}{\sqrt{2mE}}$$

de Broglie matter wave relation

$$\lambda = \frac{1.227}{\sqrt{V}} \text{ nm}$$

de Broglie wavelength of an electron accelerated through V volts

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Increasing INTENSITY increases the number of emitted electrons (current) but leaves maximum kinetic energy UNCHANGED! Only increasing FREQUENCY increases K max.

• **Alert:** [Valuation Rubric] State K max = h ν - Φ_0 = h(ν - ν_0) explicitly .

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Photoelectric effect: definition, Hertz, Hallwachs and Lenard's observations [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Photoelectric Effect Definition]** Phenomenon of emission of electrons from a metallic surface when electromagnetic radiation of sufficiently high frequency (greater than threshold frequency ν_0) strikes it. Emitted electrons are called photoelectrons. [1 Mark]

• **[Key Historical Observations]** 1. Hertz (1887): Discovered sparks jumped more readily across spark gaps when illuminated with UV light. 2. Hallwachs and Lenard: Observed negatively charged zinc plate lost its charge when exposed to UV light, but uncharged or positively charged plate was unaffected. Confirmed emitted carriers are negative electrons. [1 Mark]

• **Key Exam Result:** Photoelectric effect: UV light ejects electrons from metal surface

2. Experimental study of photoelectric effect: effect of intensity, frequency, potential on photocurrent [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Experimental Observations]** 1. Effect of Intensity: Photocurrent is directly proportional to intensity of incident radiation (at fixed frequency above ν_0). [1 Mark]

• 2. Effect of Potential: Current increases with collector potential to a constant saturation value. When collector is made negative, current decreases and drops to zero at Stopping Potential V_0. [1 Mark]

• 3. Effect of Frequency: Stopping potential V_0 increases linearly with frequency of incident light. Higher frequency implies higher maximum kinetic energy of photoelectrons. [1 Mark]

• **Key Exam Result:** Current proportional to Intensity; K max proportional to Frequency; Instantaneous emission (< 10^{-9} s)

3. Stopping potential and cutoff frequency definitions [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Definitions]** 1. Stopping Potential (Cutoff Potential, V_0): The minimum negative (retarding) potential applied to the collector plate with respect to the emitter plate at which photocurrent becomes identically zero: K max = e V_0 = (half (0.5)) m v max^2. [1 Mark]

• 2. Threshold Frequency (Cutoff Frequency, ν_0): The minimum frequency of incident electromagnetic radiation below which no photoelectric emission can take place, no matter how high the intensity of radiation is. Characteristic of the metal surface (Work function $\Phi_0 = h \nu_0$). [1 Mark]

• **Key Exam Result:** Stopping potential: e V_0 = K max | Threshold frequency: $\Phi_0 = h \nu_0$

4. Failure of wave theory of light to explain photoelectric effect [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Three Failures of Wave Theory]** 1. Kinetic Energy vs Intensity: Wave theory predicts wave energy increases with intensity, so electron KE should increase with intensity. In reality, KE depends strictly on frequency and is independent of intensity. [1 Mark]

• 2. Threshold Frequency: Wave theory predicts radiation of ANY frequency should eject electrons if intensity is high enough. In reality, no emission occurs below ν_0 regardless of intensity. [1 Mark]

• 3. Instantaneous Emission: Wave theory requires energy to be distributed over millions of surface atoms, predicting a time delay of hours before an electron absorbs enough energy. Experimentally, emission is instantaneous (< 10^{-9} s). [1 Mark]

• **Key Exam Result:** Wave theory fails: couldn't explain threshold frequency, intensity independence of KE, or zero time lag

5. Einstein's photoelectric equation h\nu = \phi 0 + K {max} = h\nu 0 + e V 0 [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Einstein's Photon Postulate & Equation]** Radiation consists of localized quanta of energy called photons: E = h ν. When a photon strikes an electron in metal, it transfers its entire energy hν to the electron. Part of energy is used to overcome binding work function Φ_0; remainder appears as maximum kinetic energy K max. [1.5 Marks]

• **[Mathematical Formulation]** h ν = Φ_0 + K max. Since $\Phi_0 = h \nu_0$: K max = h ν - h ν_0 = h (ν - ν_0). In terms of stopping potential (K max = e V_0): e V_0 = h ν - Φ_0 implies V_0 = (h divided by e) ν - (Φ_0 divided by e). Slope of V_0 versus ν graph is h divided by e (universal constant)! [1.5 Marks]

• **Key Exam Result:** h ν = Φ_0 + K max = h ν_0 + e V_0; Linear V_0 vs ν with slope h divided by e

6. Numerical on Einstein's equation: finding threshold wavelength, work function, and stopping potential [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Given Data & Formulas]** Light of wavelength $\lambda = 400\ nm = 4.0 \times 10^{-7}$ m incident on metal of work function $\Phi_0 = 2.14$ eV. [1 Mark]

• **[Step 2: Photon Energy & Calculations]** 1. Photon energy: E = h c divided by λ = ($6.626 \times 10^{-34} \times 3.0 \times 10^{8}$) divided by ($4.0 \times 10^{-7} \times 1.6 \times 10^{-19}$) eV = 1242 eV nm divided by 400 nm = 3.105 eV. 2. Maximum kinetic energy: K max = E - Φ_0 = 3.105 - 2.14 = 0.965 eV = 1.54×10^{-19} J. 3. Stopping potential: e V_0 = K max $\Rightarrow$ V_0 = 0.965 V. 4. Maximum speed: (half (0.5)) m v^2 = 1.54×10^{-19} J $\Rightarrow$ v max = $\sqrt{}$[2(1.54×10^{-19}) divided by (9.1×10^{-31})] = 5.82×10^{5} m/s. [2 Marks]

• **Key Exam Result:** E = 3.11 eV; K max = 0.965 eV; V_0 = 0.965 V; v max = 5.82×10^{5} m/s

7. Numerical on de Broglie wavelength of electron, proton, and alpha particle [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Formula for Particles]** λ = h divided by sqrt(2 m q V). [1 Mark]

• **[Comparison for Proton and Alpha Particle]** Accelerated through same potential difference V. For proton: mass = m p, charge = e. λ p = h divided by sqrt(2 m p e V). For alpha particle: mass = 4 m p, charge = 2e. $\lambda\alpha$ = h divided by sqrt(2(4 m p)(2e) V) = h divided by [sqrt(8) sqrt(2 m p e V)] = λ p divided by sqrt(8) = λ p divided by (2sqrt(2)). Ratio: λ p divided by $\lambda\alpha$ = 2sqrt(2) approx. 2.828. Proton has ~2.83 times larger de Broglie wavelength! [2 Marks]

• **Key Exam Result:** Ratio λ proton divided by λ alpha = 2sqrt(2) = 2.83 (same accelerating potential V)